Sentiment Intelligence

Bidirectional LSTM classification for short customer reviews

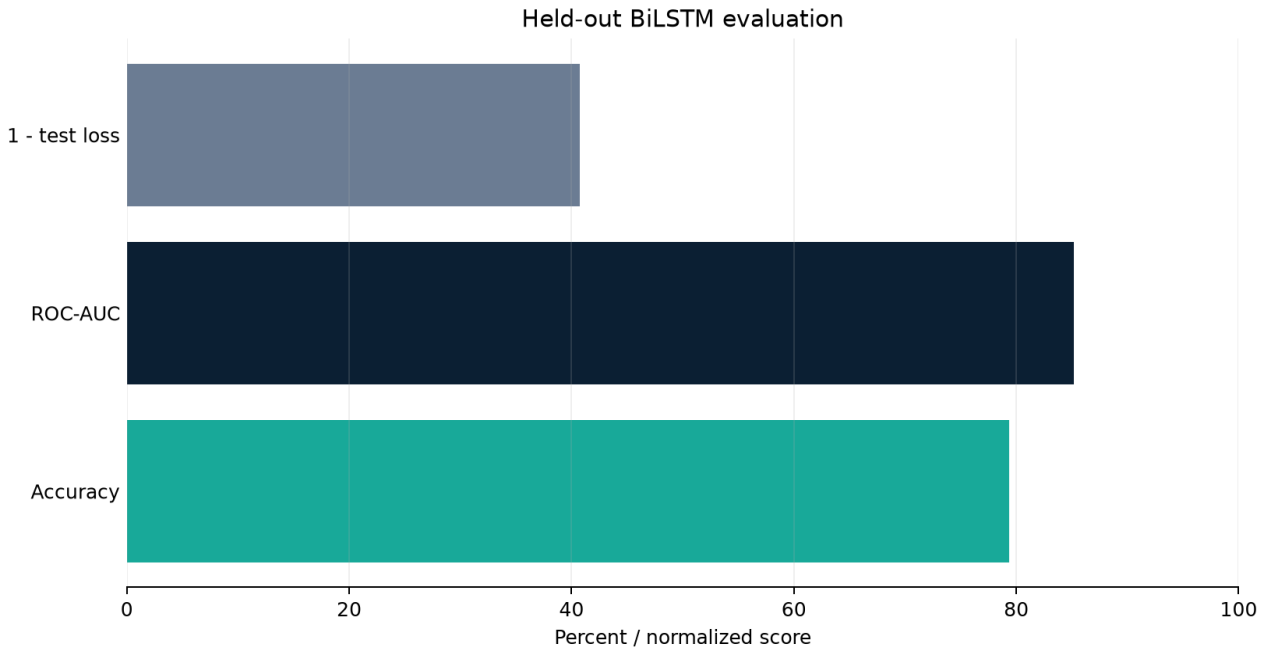
Portfolio Case Study

Executive Summary

The completed experiment combined 3,000 labeled Amazon, IMDb, and Yelp sentences and trained a compact bidirectional LSTM. The restored best model achieved 79.33% held-out accuracy, 0.8518 ROC-AUC, and 0.5925 binary cross-entropy loss. The result supports controlled sentiment triage with human review, while the observed validation divergence requires monitoring and broader validation before production use.

Decision Question

Can a compact neural network classify short customer-review sentences as positive or negative across multiple public sources?



Methods

- Loaded and validated three balanced public labeled-sentence sources.
- Cleaned text, fitted the tokenizer on training data only, and used an out-of-vocabulary token.
- Selected a 26-token sequence length from the training distribution and trained a 362,625-parameter BiLSTM.
- Evaluated on a held-out test set with accuracy, ROC-AUC, loss, and class-specific diagnostics.

Evidence and Findings

Evidence	Observed result
Corpus	3,000 labeled sentences with an even 1,500/1,500 class balance.
Held-out accuracy	79.33% using the restored best model.
Discrimination	0.8518 ROC-AUC on the held-out test set.
Generalization signal	Validation performance diverged after early epochs, indicating overfitting risk.

Limitations

- The benchmark corpus is small and may not represent current organizational language.
- Short review sentences omit conversation context, sarcasm, and domain-specific intent.
- The model requires class-specific recall and drift monitoring, not accuracy alone.

Recommended Next Actions

- Use predictions to prioritize a human-review queue rather than automate customer decisions.
- Validate on current domain text and monitor low-confidence examples and class-specific recall.
- Retrain only after collecting labeled outcomes under a documented evaluation protocol.

Reproducibility

The companion repository contains typed Python modules, tests, dependency declarations, data-access notes, and a rendered analysis notebook. Restricted source data and the working notebook are intentionally excluded from the public repository.

Results are reported as historical analytical evidence. They should be reproduced on an authorized equivalent dataset before operational use.